

WORK PROCEDURE

PRO-OP-ITH-005A

DRILLING WITH V-30 HEAD TO A BLIND BORING DOWN STOPE

OBJECTIVE

The objective of this procedure is to define safe working methods for drilling with the V-30 head to a blind boring down stope.

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL.

INTERPRETATION

Personnel with the authority for the interpretation of this document is the Operations Manager. Any request for amendment, revision or update must be made with this person.

RESPONSIBILITIES

The worker responsible for the drilling must ensure to follow these steps and follow the instructions that come from the engineering department.

METHOD

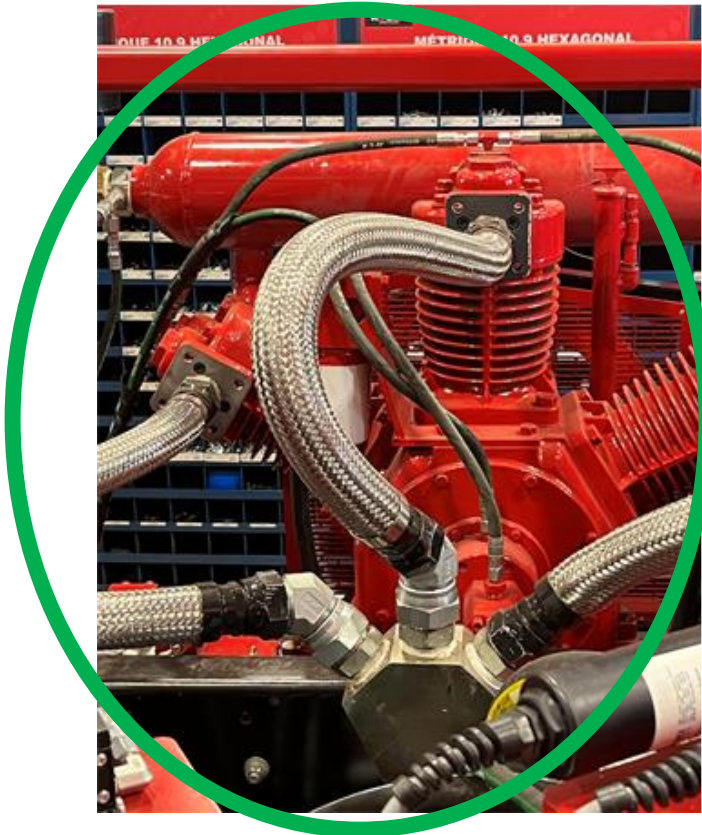
- 1)** MRI employees are authorised to move the drilling equipment into the workplace using the company's equipment provided all company regulations and directives are followed respecting the company's policies.
- 2)** A visual inspection of the unit is completed once the unit arrives at the workplace. A visual inspection is also completed of ventilation, power supply and air and water distribution.
- 3)** The electric cable is unloaded, inspected and hooked up to the ground fault distribution box. (a company electrician is required just in case a motor rotation re-connect is required)

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- 4) Verification of all control levers is required prior to moving the unit. They must be in the neutral position before powering up the unit.
- 5) A visual inspection of the unit is required for oil levels, gauges, leaks, etc. prior to moving the unit.

INSPECTIONS

It is essential to carry out a rigorous inspection of our equipment as soon as we have to operate it. In the case of CUBEX and boosters, particular attention must be paid to the high-temperature (800 degrees F) braided hoses of the compressors. For safety reasons, only the type of hose as shown below should be used.



- 6) Before the unit is moved to the first line to be drilled, with respect to the plans and specifications, a wash down of the area, a check scale and an inspection for explosives is performed in the drilling area.

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- 7) When the drill is moved in the drilling area, we take precautions to avoid damage to the electrical cable connected to the drill and check if there are no leaks or abnormalities from the drill in motion. If the extension should be handled when it is turned on, you must use the plastic hook. Barricades or sign are installed to inform all workers around about ongoing operations (Drilling, Open Hole, etc) A distance of at least 3 metres (10 feet) must be left between the barricade at the site entrance and the first pieces of equipment (toolbox, oil, etc.).

When the 600V cable is energized and it has to be handled, use the plastic hook. Install barricades or signs to advise workers about operations are ongoing.

- 8) After reaching the drilling point, the drill is turned off by the **E-STOP** and the electric cable is then hung to the wall at chest height (4 to 5 feet off the bench) to eliminate, as much as possible, any contact by loose or motor vehicles. Also, as the cable is being hung, a visual inspection of the cable is done.



The use of the E-Stop to stop the drill will result of cutting the 600 V current on all extensions.

- 9) Connecting the low-pressure air hose (2½ in. dia.) from the mine feeder to the drill booster compressor is the next priority. The supply hose is connected to the purge device, and it is blown. Then remove the hose and connect it to the air inlet of the drill. The retaining devices (whip check) are installed at each joint (if there is one). At this point, an inspection of the retainers is important. Check the condition of the wire rope and the proper functioning of the retaining mechanisms;
- 10) The re-enforced high-pressure hose (2 ½) is inspected, blown and connected from the compressor to the drill unit using whip checks at all the connections. (Note: Stem nuts on the 2 ½ in. dia. hoses are designed so that they cannot be interchanged between high and low pressure so as to eliminate any chance of a potential accident.)

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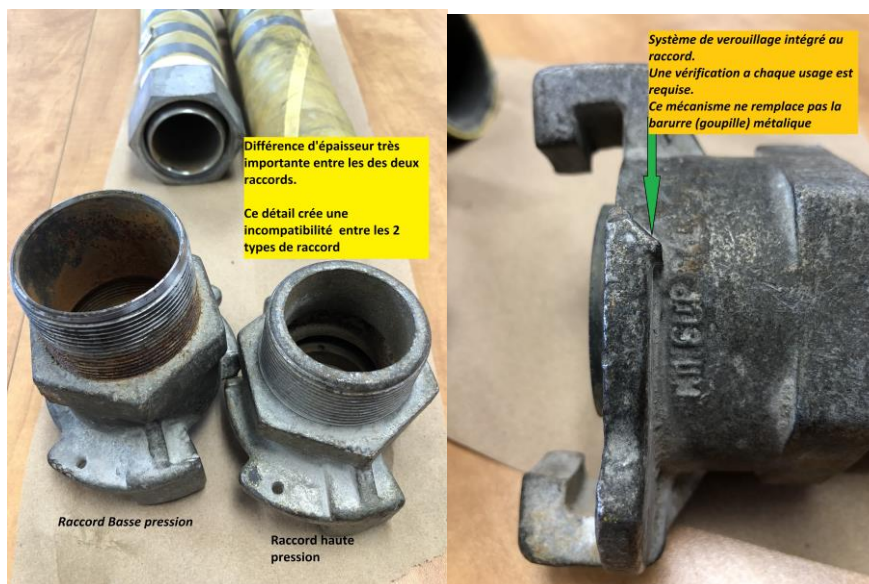


Special attention required

Connections and hoses should be thoroughly inspected to ensure that the collars are tight, that the threads of the couplings are in good condition, and that the hoses are not faulty

Difference in thickness between the two connections. This detail creates an incompatibility

Locking system integrated into the fittings. A check for each use is required. This mechanism does not replace the metal pin



- 11) The 1 in. dia. water hose is inspected, blown and connected from the mine supply to the drill.
- 12) All hoses are hung under the electric cable to eliminate any contact with loose or mine equipment.
- 13) If it is not physically possible to hang the cable and hoses (ex. due to stope dimensions) they are laid out in such a fashion to eliminate contact or crushing. They may be raised off the bench using drums or wooden tri-pods and marked (painted or identified with flagging) so that they will be visible.
- 14) Air and water lines are then slowly energized and inspected for leaks.

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- 15) The drill consol (controls) must be positioned at 1,5 meter from the mast and in such a way as to offer protection to the workers from any projection of rocks during the drilling operations;
- 16) The drill is then positioned above the drilling target (on the line and with the desired dip) as specified in the plans and specifications provided, using a magnetic laser and a digital laser clinometer for quality control;
- 17) When moving the rotating mast, special attention must be paid to the hydraulic hoses, they could get stuck. If so, go back and move the mast to loosen them. If an effort is necessary to loosen the hoses, make sure you are in a stable position ;
- 18) The hammer must be handled using the transport carriage, the bell and the chain or, at the last resort, with the assistance of a helper. **Make sure the hand truck is stable before removing the hammer on.** Bits changes and rods handling can be handled by the operator only, depending on the length of holes to be drilled.

We encourage the presence of a second worker for drilling and tearing rods for holes 200 feet and over. If it is not possible to have this 2nd worker, the driller must provide breaks

- 19) During the drilling of the hole, no one (employee or other) is authorized to circulate in front of the drill;
- 20) To install or remove the drill rods, the gear box must be raised and stopped. The drill must be in the neutral (dead man's) position before the drill rods are manipulated;
- 21) When the 6½" pilot hole is completed, the hole must be reamed to 10 in. for the centralizer (nose) of the V-30. The same method is used for installing and uninstalling the rods;
- 22) **When a break through holes is done, it is important to push, WITHOUT ROTATION, 1 rod longer than the length of the break through;**

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INSTALLATION OF THE V-30 ON THE BOOM OF THE DRILL

NB. Never stand in the axis of possible movement of the head if it is unhooked accidentally.

Make sure the controls are on neutral position

1. Position the drill in the center of the V-30 at a distance of approximately 15 feet. The head should be oriented as follows; stabilizer towards the drill and the centralizer pointing forward
2. Tilt the boom at an angle of about 60 degrees to the rear, keeping the base of the boom on the ground. If it is not possible to support the base on the ground to approach the head, extend the stinger to create a support and stabilize the mast and the drill.
3. Install the male adapter in the gear box as well as the female adapter on the drive shaft of the V-30
4. Install the chain (grade 80, 12 feet) between the gear box of the drill and the drive shaft of the V-30.
5. Use the feed cylinder to hoist the V-30 to the table until the drive shaft has entered the table enough to couple the gear box to the drive shaft of the V-30
6. Close the centralizers on the drive shaft of the V-30
7. Shut the drill OFF and remove the adapters
8. Position the mast to align the drive shaft and the gear box
9. Approach the gear box on the drive shaft and start the rotation to screw them together.
10. Put the slip plate in the slot of the rod to tighten the assembly
11. Remove the pine and open the tables
12. Fully hoist the V-30 onto the boom
13. Close the tables on each side of the V-30 to stabilize it
14. Realign the mast into the initial drilling position.
15. Align the nose of the V-30 in the pilot hole and begin to collaring the hole;

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16. As soon as the head of the V-30 has completely entered the rock (approx. 6 ft.) the drill is stopped and a cover (removable floor) is installed to provide maximum protection for employees and equipment.

Wood cover



Plastic cover



The installation of the cover provided by M.R.I. or customer on raise is mandatory at all times when drilling V-30 down;

If the cover must be removed or replaced, employees working near the raise must be attached at all times by means of an adjustable restrictive lanyard which prevents access to the hole or with a harness and a retractable block.

All fall protection equipment must be approved by the customer.

17. During drilling, if the pilot hole becomes blocked, the V-30 drill head must be removed to unblock the pilot:

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- a. When the head is out and stabilized inside the mast with the tables closed you must replace the protective cover over the V-30 opened hole and finally move the mast. Make sure that this operation can be done without one of the components of the mast or the V-30 coming into contact with the rock wall or the collar of the hole;
 - b. Remove the V-30 from the drill and place it at a distance preventing it from interfering with unblocking operations;**
 - c. The helper can guide the operator while avoiding exposure in the area of moving parts;
 - d. Reposition the mast to install the 6 inches hammer;
 - e. To unblock the pilot hole, align the 6-inches hammer in the pilot until the hole is unblocked;
18. Depending on ground conditions, changing drill bits and visual inspection are required periodically. The V-30 head is raised out and before any maintenance is performed, the cover must be installed over the hole. This is necessary for the protection of employees when performing maintenance on the head or changing bits;
19. The head is inspected for any possible repairs and then the bits can be changed at this time;
20. Sharpening and reshaping of drill bits can be done in the field by applying PRO-SEC-010 - Procedure for sharpening drill bits underground
21. If V-30 raises are planned for more than 40 feet of drilling, the M.R.I. driller must install a centralizer to stabilize the rods;

IT IS STRICTLY FORBIDDEN FOR ANYONE TO ENTER THE 30-INCH RAISE TO RECOVER ANYTHING, REGARDLESS TO HIS DEPTH

M.R.I. HAS SUITABLE EQUIPMENTS AND TOOLS TO EXTRACT MATERIALS THAT ARE BROKEN IN THE HOLES (EX. DRILL RODS OR DRILL BITS) TO KEEP DRILLING

22. The steel cover must be installed:
- a. As soon as V-30 drilling is completed, OR;
 - b. That its drilling is suspended to drill production holes, OR;
 - c. To perform unblocking;

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Steel cover



23. The cover must remain on the raise until the mine blasting crew removes it;
24. When the equipment is removed from the drill site, a barricade and/or signs must be installed so that all persons are notified that there is an open hole condition in the area. The customer is informed of the situation and can take the necessary security precautions that apply;

Procedure for dealing with rod failure:

- a) If a rod failure occur, immediately stop drilling, shut down electrical power and air supply to the drill.
- b) A barricade or the necessary signs must be posted at the entrance to the work area in such a manner as to close off the work area and preventing any access to the site.
- c) The driller must notify his work supervisor immediately and no work will resume until the supervisor is on site.
- d) The driller assisted by the supervisor will assess the situation and determine the means by which the failed rod or rods will be removed and replaced.
- e) The site supervisor will remain on site throughout the recovery process.
- f) All drill operation required to remediate the situation will be carried out using the remote control consol. The consol will be located in area removed from the drill in order to ensure that employees are protected from any potential falling objects.
- g) Once the failed rods have been removed and replaced, all remaining rods in the hole will be removed in order to ensure no other rods in the drill string have been damaged.

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- h) The site supervisor will determine that the equipment is in the condition required to resume drilling operations & will give the sign off for drilling to resume.
- i) The site supervisor will have the failed rods tagged & returned to the vendor in order to determine the likely cause of the failure;
- j) The site supervisor will be responsible for completing an incident report that will detail the rod failure event, the recovery operation and resulting corrective actions.

Ajouts ou modifications	Date	Par
Added : Obligation to install the metal cover over the V-30 hole at point #28	December 3rd 2020	S. Tremblay
Change the phrasing in point #29 for "IT IS STRICTLY FORBIDDEN"	Decembre 3rd 2020	S. Tremblay
Add point #17 of the possibility that the hydraulic hoses can be stuck and the measures to be taken if this occurs.	12-01-2023	S. Tremblay
Complete revision	03-2023	S. Tremblay JF Gagnon S. Desrosiers
Added warning on inspections and high-temperature hoses	7-04-2023	S. Tremblay
Addition of a 3-meter distance between the ribbon and the first pieces of equipment in the work area. Addition made to point #7 and on the " STANDARD INSTALLATIONS" following the Amaruk event.	19-12-2023	S. Tremblay S. Desrosiers
Translation revision	23-04-2024	S. Tremblay D. Baribeault
Add #22 pushing one more rod than the length of the breakthrough	21-06-2024	S. Tremblay JF Gagnon



Christian St-Amour
Operation Manager

Standard installations (not scale)

Keep 7 meters between Cubex and booster

A distance of at least 3 metres (10 feet) must be left between the barricade at the site entrance and the first pieces of equipment (toolbox, oil, etc.).

